Armadale Kalamunda Group

Prevention of Perioperative Hypothermia Guideline

1. Background

This guideline is relevant to patients undergoing a procedure or surgery in one of the AKG theatres or procedure rooms

2. Purpose

To improve awareness and highlight the importance of maintaining normothermia during perioperative period among patients, carers, health care workers and supporting staff.

To implement standardized practice to prevent inadvertent perioperative hypothermia.

To avoid preventable adverse outcomes, improve patient care and satisfaction.

3. Applicability

This guideline is relevant not only to patients undergoing a procedure or surgery in one of the AKG theatres or procedure rooms, but it applies to their carers as well as all medical and non-medical staff involved in perioperative care. This includes all healthcare professionals involved in the patient journey: preadmission clinic, same day, and admission wards, holding bay, operating theatres, PACU and postoperative destination ward. In addition, all staff working in the emergency department, intensive care unit, surgical, medical, obstetric, post-natal, paediatric, psychiatry, rehab and 23-hour ward and supporting staff who care for these patients during the perioperative period are all included. The guideline is not intended for patients with hyperthermia, patients undergoing therapeutic hypothermia or patients with severe head injuries resulting in impaired temperature control.

4. Guidelines

4.1 Inadvertent perioperative hypothermia

Inadvertent perioperative hypothermia is an important health indicator. It is collected and reported by the Australian Council of Healthcare Standards (ACHS) which forms the basis of The National Safety and Quality Health Service (NSQHS) Standards. Out of 18 clinical indicators reported for Anaesthesia and Perioperative Care sector, Hypothermia (Temperature less than 36’C in the recovery period) is one of the important indicators. In the latest report (21st edition; 2012-2019), the average hypothermia rate was 1.5 % (higher than 10% in some outlier hospitals). From 2014 to 2018, the annual average rate was between 2.5 to 1.9 %.

The AHS hypothermia rate was 7% in 2014. Theatre staff education programs, peer information and CME sessions, gradual change in the practice and annual auditing have resulted in significant reduction of AHS hypothermia rate from 7% in 2014 to less than 1% since 2018 (for non-endoscopic procedures; since 2018, the endoscopy data are not represented in TMS). The current guideline aims to maintain the result, formalise the awareness and standardise the practice.

4.2 Physiology of perioperative hypothermia and its implications

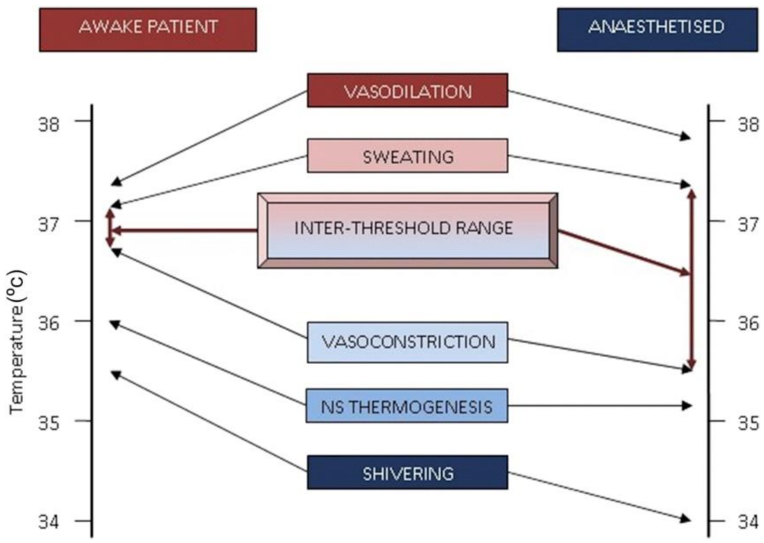
4.2.1 Normal thermoregulation

Body temperature is tightly regulated in the core compartment (mainly in the trunk and head), which constitutes approximately one-half of the body mass and directs central thermoregulatory responses. Peripheral tissues (the arms and legs) act as a thermal buffer and vary over a wide range. They are 2 to 4°C cooler than the core temperature in patients exposed to hospital ambient temperatures.

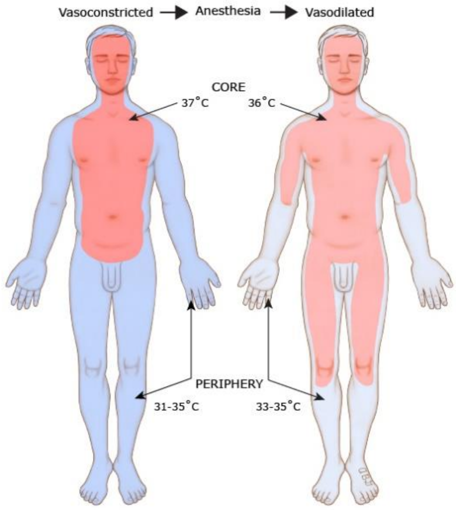
Normally, the posterior hypothalamus maintains a temperature set point between 36.7 and 37.1’ C. Outside this threshold range, the hypothalamus initiates homeostatic and behavioural mechanisms to return the body to normothermia.

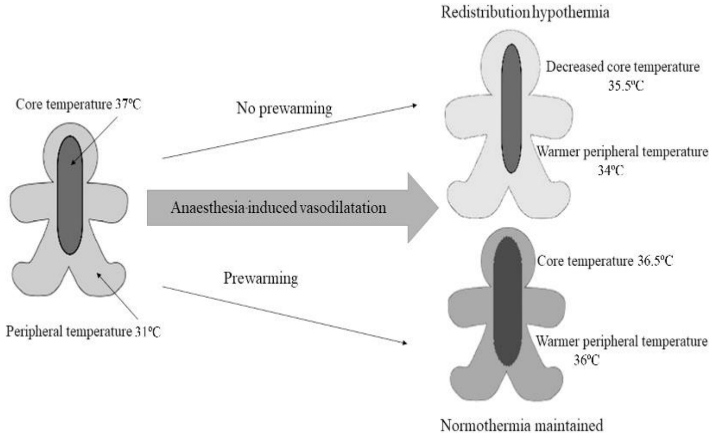
The primary human thermoregulatory defences in response to cold are shivering and peripheral arteriovenous shunt vasoconstriction that regulate arm and leg blood flow and, thus, heat loss to the environment.

However, intraoperative models of the human thermoregulatory system are not as simple as an "on/off" thermostat because of sequential activation of defences as well as the effects of anaesthesia on thermoregulatory control.



In older adults, cold-defence thresholds are reduced approximately 1°C. In infants, but not in adults, non-shivering thermogenesis is an additional thermoregulatory defence.





4.2.2 Effects of general anaesthesia

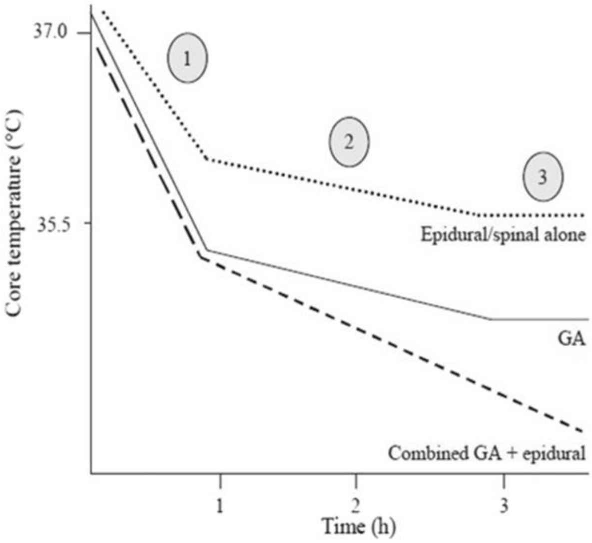
Volatile agents (sevoflurane, desflurane as well as nitrous oxide), intravenous (IV) anaesthetic agents (propofol, dexmedetomidine) and opioids impair thermoregulatory control in a dose-dependent fashion. Agents used to maintain general anaesthesia synchronously reduce the thresholds for vasoconstriction and shivering in a concentration dependent manner (linear for IV anaesthetic agents, but disproportionately greater at higher concentrations of volatile anaesthetics). Shivering is not possible when patients are paralysed. The result is that the normally narrow regulated range of core temperatures increases 10- to 20-fold during general anaesthesia.

4.2.3 Effects of neuraxial or regional anaesthesia

The patients receiving neuraxial anaesthesia become hypothermic to a degree similar to those having general anaesthesia. Impairment of thermoregulatory control by neuraxial anaesthesia occurs via three mechanisms:

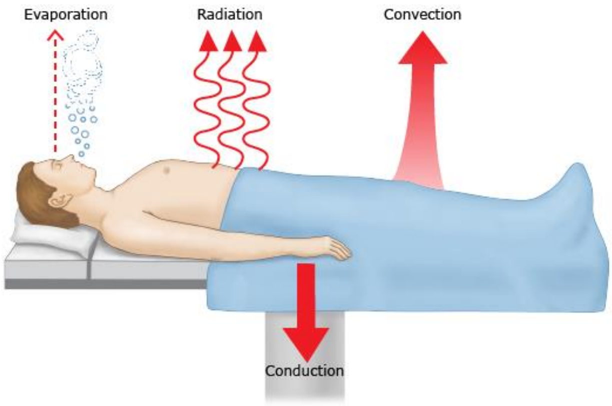
- Hypothermia incites less thermal discomfort. Patients do not feel cold during epidural or spinal anaesthesia even if they are hypothermic. The tonic cold signals from the lower body are blocked and this is interpreted by the hypothalamus as indicating a relatively warm state.
- Neuraxial anaesthesia impairs central thermoregulatory control, reducing the vasoconstriction and shivering thresholds. Changes in the shivering threshold are proportional to the block height. However, impairment of central thermoregulatory control is less after neuraxial anaesthesia compared with general anaesthesia.
- Neuraxial anaesthesia prevents most efferent and afferent autonomic neural activity to and from the lower body to the brain including afferent pain signals and the efferent signals controlling vasoconstriction and shivering. With impaired neurally mediated signalling, the gain and maximum intensity of thermoregulatory defences are substantially reduced.

4.2.4 Phases of intraoperative hypothermia



- In the first hour after induction of general anaesthesia or activation of a neuraxial block, core temperature decreases rapidly due to redistribution of heat from the core to the periphery. This is due to anaesthetic-induced vasodilation resulting from impairment of central thermoregulatory control rather than direct peripheral effects of anaesthetics. Although this redistribution of heat does not alter mean body temperature, it significantly reduces core temperature.
- Redistribution hypothermia is followed by a slower, linear reduction in core temperature resulting from heat loss exceeding metabolic heat production. The main routes of heat loss are usually radiation and convection (only small amounts are lost via conduction and evaporation). Heat is lost through surgical incisions, substantial when incisions are large, and especially during open abdominal surgery. The rate of decrease is a function of the difference between metabolic heat production and heat loss, which in turn depends on ambient temperature, extent of surgical exposure of skin or body cavities, and whether the patient is insulated or actively warmed.
- Once patients become sufficiently hypothermic to activate thermoregulatory vasoconstriction (approximately 34.5°C during general anaesthesia), core temperature reaches a plateau and does not decrease further regardless of the incision size or duration of surgery. It also results from a passive process if heat production is balanced against heat loss. Once activated, arteriovenous shunt vasoconstriction effectively retains metabolic heat in the core tissues, thus preventing further decreases in core hypothermia. However, heat loss from peripheral tissues continues. Consequently, body heat content continues to decrease even though core temperature remains constant.

4.3 Mechanism of heat loss



4.3.1 Radiation (40%)

Thermal radiation acts like the electromagnetic spectrum; it travels in lines and can be reflected. Heat loss is proportional to the temperature difference to the power of 4.

4.3.2 Convection (30%)

Caused by a movement away from a heat source by molecules that have increased velocity because of transfer of heat energy. Heat loss is proportional to the velocity of air currents around the body and is exacerbated in the operating theatre by air currents and laminar flow.

4.3.3 Evaporation (25%)

Heat is lost by latent heat of vaporisation, where heat from the body is used to turn water into vapor. Examples include sweating or evaporation from exposed body cavities or mucosal surfaces, skin preparation fluids, and the humidification of any dry respiratory gases.

4.3.4 Conduction (5%)

Transfer of heat from a warm object in direct contact with a cooler object, for example, the use of non-warmed IV fluids or contact with a cold theatre table or mattress.

4.4 Complications and consequences of perioperative hypothermia

Most cellular functions are temperature-dependent, and hypothermia also provokes systemic responses. Thus, even mild hypothermia may have adverse consequences.

4.4.1 Coagulopathy

Hypothermia causes reversible impairment of platelet aggregation via reduced release of thromboxane A3, which impairs formation of an initial platelet plug. It also reduces the activity of enzymes in the coagulation cascade, which in turn reduces clot formation.

This combination increases perioperative blood loss and the need for transfusion. (Increase relative risk by 22%). Hypothermia-induced coagulopathy is often missed with routine laboratory testing if they are not temperature-adjusted.

4.4.2 Infection

There are three mechanisms by which perioperative hypothermia impairs host defences against surgical wound contamination:

- Reduced tissue perfusion to wounded tissue with reduced access for key immune cells
- Decreased motility of key immune cells (macrophages)
- Reduced scar formation, necessary to prevent wound dehiscence and recontamination.

4.4.3 Prolongation of drug effects

Even mild hypothermia prolongs the action of neuromuscular blocking agents (NMBAs).

Plasma propofol concentrations are increased by 28 percent with hypothermia (up to 3°C), as well as with opioids due to reduced hepatic blood flow.

Decreased minimum alveolar concentration of volatile inhalation agents.

Consequences include delayed reversal of neuromuscular blockade and delayed emergence in hypothermic patients.

4.4.4 Shivering

The shivering threshold is below the vasoconstriction threshold. Vasoconstriction is effective even during anaesthesia. Consequently, core temperature rarely decreases to the temperature necessary to reach the shivering threshold. Intraoperative shivering is rare for this reason, and because anaesthesia impairs activation of shivering.

By contrast, postoperative shivering is common in hypothermic patients when anaesthesia has worn off. Shivering should be avoided since it acutely augments metabolic rate, and patients find it uncomfortable. It increases carbon dioxide production, catecholamine release, and cardiac output. Ischaemic cardiac events can be worsened by the increased skeletal muscle oxygen demand of shivering.

4.5 Patient outcome

Hypothermia is associated with prolonged hospital stay and health care related complications.

4.5.1 Myocardial ischemia

Hypothermia results in sympathetic stimulation with augmentation of plasma norepinephrine concentration (up to 700 %) with consequent increased metabolic rate and hypertension. These effects increase myocardial oxygen consumption.

In patients with ischemic heart disease, it may lead to myocardial ischemia and arrhythmias.

4.6 Specific patient groups

4.6.1 Paediatrics

Neonates and young children lose more heat than adults via conduction and radiation due to higher surface area to volume ratio and less insulating subcutaneous adipose tissue. They have increased basal metabolic rate. Newborns have an immature hypothalamic thermoregulatory capacity and a high resting vagal tone and are unable to vasoconstrict. General anaesthesia impairs the compensatory non shivering thermogenesis.

4.6.2 Pregnant patients undergoing LSCS

The emerging evidence indicates that hypothermia can occur during LSCS in alert mothers undergoing LSCS procedure under neuraxial anaesthesia.

5. Procedure

5.1 Preoperative phase

During the preoperative interview or assessment by the pre-admission staff or anaesthetist, patients (and their families and carers) should be informed that:

- Staying warm before surgery will lower the risk of postoperative complications
- The hospital environment may be colder than their own home
- They should bring additional clothing, such as a dressing gown, a vest, warm clothing, socks, and slippers, to help them keep comfortably warm
- They should tell staff if they feel cold at any time during their hospital stay and request extra blankets if needed.

During that preop interview/ assessment high risk patients should be identified and recorded in the notes behind the Anaesthetic chart by the anaesthetist and preventive measures are taken.

The patients are identified as higher risk if any two of the following apply.

- ASA grade more than I (the higher the grade, the greater the risk)
- Preoperative temperature below 36.0°C (and preoperative warming is not possible because of clinical urgency)
- Undergoing combined general and regional anaesthesia
- Undergoing major joint arthroplasty, major or intermediate surgery or surgery time estimated for more than two hours
- Major Adverse Cardiac Event (MACE) rate estimated to be equal or more than 1 % (as per ACC/AHA Guideline) or Revised Cardiac Risk Index equal or more than 1 point.

Pay particular attention to the comfort of patients with communication difficulties before, during and after surgery.

Every patient in the admission ward or emergency department must be provided with blankets after changing into theatre attire. The patient's temperature must be measured and documented on admission, and before they leave the ward or emergency department. If the patient remains for more than 2 hrs before going to theatre, their temperature should be recorded and inform the anaesthetist if the temperature is less than 36.0 C.

If the patient's temperature is below 36.0°C, inform the relevant anaesthetist and start active warming using approved forced air warming device preoperatively on the ward or in the emergency department (unless there is a need to expedite surgery because of clinical urgency, for example bleeding or critical limb ischaemia). Forced air warming is only to be used with its approved warming blanket and never to be used directly to patients or applied between the sheets of blankets. Ideally, the patient's temperature should be 36.0°C or above before they are transferred from the ward or emergency department. On transfer to the theatre suite, active warming should be continued (or re-started as soon as possible).

Once the patient arrives in the holding bay, measure the temperature and if it is 36.0°C or below, inform the relevant anaesthetist and start active warming using forced air warming. Continue this until induction of anaesthesia, unless this will delay emergency surgery, and maintain active warming throughout the induction and intraoperative phase with regular temperature monitoring, at least every 15 minutes until normothermia is achieved.

5.2 Intraoperative phase

Patient temperature measurement and control.

The patient's temperature should be measured and documented before induction of anaesthesia and then every 30 minutes until the end of surgery.

The patient should be adequately covered throughout the intraoperative phase to conserve heat and exposed only for surgical preparation.

The temperature setting and airflow on forced-air warming devices and heating mattress should be adjusted to maintain a patient temperature of at least 36.5°C. Meticulous attention is needed to attend to alarms indicating faults and errors in device operation and frequent monitoring to make sure the patient is not overheated.

5.3 Inside the theatre

5.3.1 In the theatre suite

Ambient temperature should be at least 21°C (24°C for Paediatrics) while the patient is exposed.

Once active warming is established, the ambient temperature can be adjusted to allow better working conditions for surgical team

Using equipment to cool the surgical team should also be considered.

5.3.2 Induction of anaesthesia

Induction of anaesthesia should not begin unless the patient's temperature is 36.0°C or above except clinical and obstetric emergency including bleeding or limb ischaemia

Warm patients intraoperatively from induction of anaesthesia, using a forced-air warming device and/or heating mattress.

Consider a resistive heating mattress if a forced-air warming device is unsuitable or is unable to cover the desired surface area. Half body heating mattresses are considered for lithotomy position in addition to forced-air warming devices.

5.3.3 Fluid warming

Blood and blood products should be warmed to 37°C using approved fluid warming devices.

All IV maintenance fluid and irrigation fluids used intraoperatively should be warmed in a thermostatically controlled cabinet to a temperature of 38°C to 40°C.

5.4 Postoperative phase

5.4.1 In recovery

The patient's temperature should be measured and documented on admission to the recovery room and then every 15 minutes.

Ward transfer should not be arranged unless the patient's temperature is 36.0°C or above.

If the patient's temperature is below 36.0°C, they should be actively warmed using forced-air warming until they are discharged from the recovery room or until they are comfortably warm.

5.4.2 On the ward

Patients should be kept comfortably warm when back on the ward.

Their temperature should be measured and documented on arrival to the ward.

Temperature should then be measured and documented as part of routine observations.

They should be provided with at least 1 cotton sheet plus 2 blankets, or a duvet.

If the patient's temperature falls below 36.0°C while on the ward:

- They should be warmed using forced-air warming until they are comfortably warm.
- Temperature should be measured and documented at least every 30 minutes during warming.

5.5 Safety, quality, and standards

5.5.1 Thermometry

The most widely used device in AKG theatres are infra-red tympanic thermometers. They are not accurate to 0.5°C and are also user dependent and prone to operator error. Aural readings are affected by ear debris and sometimes ear canal temperature is measured instead of tympanic membrane. Low readings can be compared with the opposite ear and if in doubt, oral (sublingual) temperature may need to be taken if it is safe to measure without harm to the patient.

For higher risk patients and prolonged surgery under general anaesthesia, intraop nasopharyngeal (inserted up to 10cm) or oesophageal (inserted up to 40cm) continuous temperature monitoring may be needed. This provides 0.5°C accuracy of core temperature.

5.5.2 Heating devices

The health care professionals required to use force air warming devices and electronic heat mattress should undergo appropriate training and must be competent in using these devices.

Temperature monitoring and heating devices should be checked daily and serviced regularly as per manufacturer and site-specific recommendations.

Thermal injury and burns have been reported by inappropriate usage of devices such as using force air warmer directly, between the sheets or blankets and breakage in the internal connection inside the heating mattress causing burns. They are due to inappropriate handling and storage such as folding and crushing the mattresses.

There are also studies reporting association of surgical site infection and incorrect usage of force air warming devices but there is no significant association between appropriate use and adverse outcomes. As heating devices can lead to hyperthermia if used for long periods of time without frequent monitoring or attention, it is advisable to monitor temperature every 15 – 30 minutes or in case of continuous temperature monitoring, setting appropriate alarm limits before the start of the case followed by vigilant monitoring.

6. Compliance monitoring

The new guideline must be introduced to all stake holders and relevant staff. A questionnaire including feedback request will be sent out to all the staff to confirm the correct understanding and awareness. Feedback will be reviewed and will be used for further improvement of the process. Targeted and regular audits must be conducted, and results are to be discussed with relevant teams to reinforce areas which need special attention and further improvement.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 – Action 1.01
- Standard 2 – Action 2.01

National Standards for Mental Health Services (2010):

- Standard 1 – Criteria 1.1

Australia/New Zealand Standards™ (AS/NZS):

- AS/NZS 0000:0000

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9. Glossary

The following definitions are relevant to this guideline.

Term	Definition
Hypothermia	Hypothermia is defined as a core body temperature less than 36.0°C.
Normothermia	Normothermia is defined as a core temperature range of 36.5°C to 37.5°C.
Hyperthermia	Hyperthermia is defined as a core temperature greater than 37.5°C.
Core Temperature	Core temperature is the temperature of the blood and internal organs.
Perioperative Period	The patient care pathway initiates from preadmission clinic interviews (phone, telehealth or in person) followed by pre, intra and post-operative phases. In emergency cases, it starts from the time a patient is decided for surgical intervention by the treating team.
Preoperative phase	Defined as the period from the time of preparation for surgery/administration of premedication to the time of first anaesthetic intervention.
Intraoperative phase	Defined as the period from time of anaesthetic intervention to entry into the recovery room.
Postoperative phase	Up to but not limited to 24 hours postoperatively, commencing from transfer to the recovery room and including the postoperative destination clinical area (e.g., ward, ICU).
Forced air warming	A temperature management unit where heated air is used to warm patients through convection. The warming unit draws ambient air through a filter and warms the air to a specified temperature. The warmed air is delivered through a hose only to a specifically designed blanket or gown and never through the layers between the blankets nor directly to the patient.
Electric warming mattress	An active patient warming device placed underneath the patient delivering warming at a low voltage (24V). A control unit is used to maintain the mattresses at the user-selected temperature.

Term	Definition
	Surfaces are anti-static, latex-free polyurethane with fully welded seams.

10. Document contact

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11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V1	04/09/2023	04/09/2026	New document

12. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Dr Alison Parr – Director Clinical Services
Authorisation date	01/09/2023
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